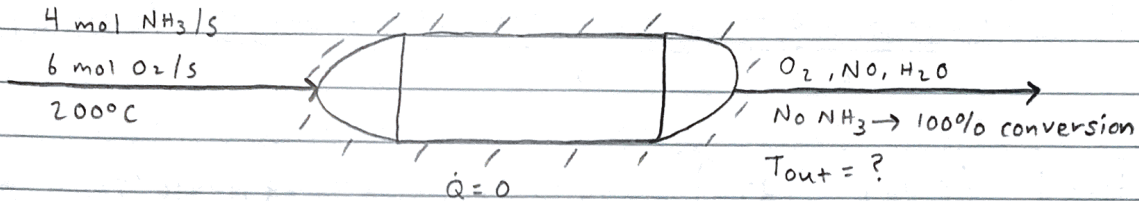


54
60

1.) Burning Ammonia



- Two solutions were found for the exit temperature $\rightarrow 2657\text{ K}$ and 3626 K
- Rework the energy balance using better expression for the heat capacities

Material Balance

$$\text{NH}_3: \dot{n}_{\text{NH}_3, \text{out}} = \dot{n}_{\text{NH}_3, \text{in}} - 4\dot{\xi} \rightarrow \dot{\xi} = \frac{1}{4} \dot{n}_{\text{NH}_3, \text{in}} = \frac{1}{4}(4 \text{ mol/s}) \rightarrow \boxed{\dot{\xi} = 1 \text{ mol/s}}$$

$$\text{O}_2: \dot{n}_{\text{O}_2, \text{out}} = \dot{n}_{\text{O}_2, \text{in}} - 5\dot{\xi} \rightarrow \dot{n}_{\text{O}_2, \text{out}} = (6 \text{ mol/s}) - 5(1 \text{ mol/s}) \rightarrow \boxed{\dot{n}_{\text{O}_2, \text{out}} = 1 \text{ mol/s}}$$

$$\text{NO}: \dot{n}_{\text{NO}, \text{out}} = \dot{n}_{\text{NO}, \text{in}} + 4\dot{\xi} \rightarrow \dot{n}_{\text{NO}, \text{out}} = 4(1 \text{ mol/s}) \rightarrow \boxed{\dot{n}_{\text{NO}, \text{out}} = 4 \text{ mol/s}}$$

$$\text{H}_2\text{O}: \dot{n}_{\text{H}_2\text{O}, \text{out}} = \dot{n}_{\text{H}_2\text{O}, \text{in}} + 6\dot{\xi} \rightarrow \dot{n}_{\text{H}_2\text{O}, \text{out}} = 6(1 \text{ mol/s}) \rightarrow \boxed{\dot{n}_{\text{H}_2\text{O}, \text{out}} = 6 \text{ mol/s}}$$

Species	$\dot{n}_{\text{in}} (\text{mol/s})$	$\hat{H}_{\text{in}} (\text{kJ/mol})$	$\dot{n}_{\text{out}} (\text{mol/s})$	$\hat{H}_{\text{out}} (\text{kJ/mol})$
$\text{NH}_3(\text{g})$	4	$\hat{H}_1$	—	—
$\text{O}_2(\text{g})$	6	$\hat{H}_2$	1	$\hat{H}_3$
$\text{NO}(\text{g})$	—	—	4	$\hat{H}_4$
$\text{H}_2\text{O}(\text{g})$	—	—	6	$\hat{H}_5$

(Murphy B.17)
$$\hat{H}_1 = \int_{298\text{K}}^{473\text{K}} (C_p(T)_{\text{NH}_3(\text{g})}) dT = \int_{298\text{K}}^{473\text{K}} (27.31 + 0.02383T + 1.707 \times 10^{-5}T^2 - 1.185 \times 10^{-8}T^3) dT$$

$$\boxed{\hat{H}_1 = 6.714 \text{ kJ/mol}}$$

$$\hat{H}_2 = \int_{298\text{K}}^{473\text{K}} (C_p(T)_{\text{O}_2(\text{g})}) dT = \int_{298\text{K}}^{473\text{K}} (28.11 - 3.68 \times 10^{-6}T + 1.746 \times 10^{-5}T^2 - 1.065 \times 10^{-8}T^3) dT$$

$$\boxed{\hat{H}_2 = 5.269 \text{ kJ/mol}}$$

$$\hat{H}_3 = \int_{298K}^T (C_p(T)_{O_2(g)}) dT = \int_{298K}^T (28.11 - 3.68 \times 10^{-6} T + 1.746 \times 10^{-5} T^2 - 1.065 \times 10^{-8} T^3) dT$$

$$((\text{in terms of } T)) \hat{H}_3 = 28.11T - \frac{(3.68 \times 10^{-6})T^2}{2} + \frac{(1.746 \times 10^{-5})T^3}{3} - \frac{(1.065 \times 10^{-8})T^4}{4} - (8514 \text{ J/mol K})$$

$$\hat{H}_4 = \int_{298K}^T (C_p(T)_{NO(g)}) dT = \int_{298K}^T (29.35 - 9.378 \times 10^{-4} T + 9.747 \times 10^{-6} T^2 - 4.187 \times 10^{-9} T^3) dT$$

$$((\text{in terms of } T)) \hat{H}_4 = 29.35T - \frac{(9.378 \times 10^{-4})T^2}{2} + \frac{(9.747 \times 10^{-6})T^3}{3} - \frac{(4.187 \times 10^{-9})T^4}{4} - (8782 \text{ J/mol K})$$

$$\hat{H}_5 = \int_{298K}^T (C_p(T)_{H_2O(g)}) dT = \int_{298K}^T (32.24 + 0.001924T + 1.055 \times 10^{-5} T^2 - 3.596 \times 10^{-9} T^3) dT$$

$$((\text{in terms of } T)) \hat{H}_5 = 32.24T + \frac{(0.001924)T^2}{2} + \frac{(1.055 \times 10^{-5})T^3}{3} - \frac{(3.596 \times 10^{-9})T^4}{4} - (9779 \text{ J/mol K})$$

Energy Balance

$$\sum_e \dot{n}_e \Delta \hat{H}_e - \sum_i \dot{n}_i \Delta \hat{H}_i + \dot{Q} \Delta \hat{H}_r^0 \quad (\Delta \hat{H}_r^0 = \sum_i \nu_i \Delta \hat{H}_{f,i}^0 = \sum_i \nu_i \Delta \hat{H}_{c,i}^0)$$

(Murphy B.3)

Species	$\Delta \hat{H}_f^0$ (kJ/mol)	$\Delta \hat{H}_c^0$ (kJ/mol)
NH ₃ (g)	-46.15	-316.8
O ₂ (g)	0	0
NO(g)	90.25	-90.2
H ₂ O(g)	-241.83	0

$$\Delta \hat{H}_r^0 = \left(4(90.25) + 6(-241.83) - 4(-46.15) - 5(0) \right) \frac{\text{kJ}}{\text{mol}} = -905.38 \text{ kJ/mol}$$

$$= \left(4(-90.2) + 6(0) - 4(-316.8) - 5(0) \right) \frac{\text{kJ}}{\text{mol}} = -906.4 \text{ kJ/mol}$$

* Using the average of both values... $\frac{-905.38 - 906.4}{2} = -905.9 \text{ kJ/mol}$

$$\Delta \hat{H}_r^0 = -905.9 \text{ kJ/mol}$$

$$\Delta \dot{H} = 0 = \sum_e \dot{n}_e \Delta \hat{H}_e - \sum_i \dot{n}_i \Delta \hat{H}_i + \dot{Q} \Delta \hat{H}_r^\circ$$

$$0 = [(1 \text{ mol/s}) \hat{H}_3 + (4 \text{ mol/s}) \hat{H}_4 + (6 \text{ mol/s}) \hat{H}_5] - [(4 \text{ mol/s}) \hat{H}_1 + (6 \text{ mol/s}) \hat{H}_2] + [(1 \text{ mol/s}) \Delta \hat{H}_r^\circ]$$

$$0 = \left[(1 \text{ mol/s}) \left(28.11T - \frac{(3.68 \times 10^{-6})T^2}{2} + \frac{(1.746 \times 10^{-5})T^3}{3} - \frac{(1.065 \times 10^{-8})T^4}{4} - 8514 \text{ J/mol K} \right) + \right.$$

$$\left[(4 \text{ mol/s}) \left(29.35T - \frac{(9.378 \times 10^{-4})T^2}{2} + \frac{(9.717 \times 10^{-6})T^3}{3} - \frac{(4.187 \times 10^{-9})T^4}{4} - 8782 \text{ J/mol K} \right) + \right.$$

$$\left. \left[(6 \text{ mol/s}) \left(32.24T + \frac{(0.001924)T^2}{2} + \frac{(1.055 \times 10^{-5})T^3}{3} - \frac{(3.96 \times 10^{-9})T^4}{4} - 9779 \text{ J/mol K} \right) \right] \right]$$

$$-58.47 \text{ kJ/s} = [(4 \text{ mol/s})(6.714 \text{ kJ/mol}) + (6 \text{ mol/s})(5.269 \text{ kJ/mol})]$$

$$-905.9 \text{ kJ/s} = [(1 \text{ mol/s})(-905.9 \text{ kJ/mol})]$$

(combining terms)

$$0 = \left[\frac{(338.95T + 3.874 \times 10^{-3}T^2 + 3.992 \times 10^{-5}T^3 - 1.200 \times 10^{-8}T^4 - 102,316)}{1000} \right] + (-964.37 \text{ kJ/s})$$

$$964.37 = (338.95T + 3.874 \times 10^{-3}T^2 + 3.992 \times 10^{-5}T^3 - 1.200 \times 10^{-8}T^4 - 102,316) / (1000)$$

$$\rightarrow T = 2657 \text{ K and } T = 3626 \text{ K} \rightarrow \text{using Murphy}$$

(NIST)

$$\hat{H}_1 = \int_{298 \text{ K}}^{473 \text{ K}} (C_p(T)_{\text{NH}_3(g)}) dT = H^\circ - H^\circ_{298.15} = At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} - \frac{E}{t} + F - H$$

$$\hat{H}_1 = 6.69866 \text{ kJ/mol}$$

$$\hat{H}_2 = \int_{298 \text{ K}}^{473 \text{ K}} (C_p(T)_{\text{O}_2(g)}) dT = H^\circ - H^\circ_{298.15} = At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} - \frac{E}{t} + F - H$$

$$\hat{H}_2 = 5.24836 \text{ kJ/mol}$$

$$\hat{H}_3 = \int_{298 \text{ K}}^{473 \text{ K}} (C_p(T)_{\text{O}_2(g)}) dT = H^\circ - H^\circ_{298.15} = \int_{298 \text{ K}}^{473 \text{ K}} \left(At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} - \frac{E}{t} + F - H \right) dT$$

$$\hat{H}_3 = 31.32234T + \frac{(-20.23531)T^2}{2} + \frac{(57.86644)T^3}{3} + \frac{(-36.50624)T^4}{4} - \frac{(-0.007374)}{T} + (-8.903471)$$

$$\hat{H}_4 = \int_{298K}^{473K} (C_p(T)_{NO(g)}) dT = \int_{298K}^{473K} \left(At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} - \frac{E}{t} + F - H \right) dT$$

$$\hat{H}_4 = 23.83491T + \frac{(12.58878)T^2}{2} + \frac{(-1.139011)T^3}{3} + \frac{(-1.497759)T^4}{4} - \frac{(0.214194)}{T} + (83.35783)$$

$$\hat{H}_5 = \int_{298K}^{473K} (C_p(T)_{H_2O(g)}) dT = \int_{298K}^{473K} \left(At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} - \frac{E}{t} + F - H \right) dT$$

$$\hat{H}_5 = 30.09200T + \frac{(6.832514)T^2}{2} + \frac{(6.793435)T^3}{3} + \frac{(-2.534480)T^4}{4} - \frac{(0.082139)}{T} + (-250.8810)$$

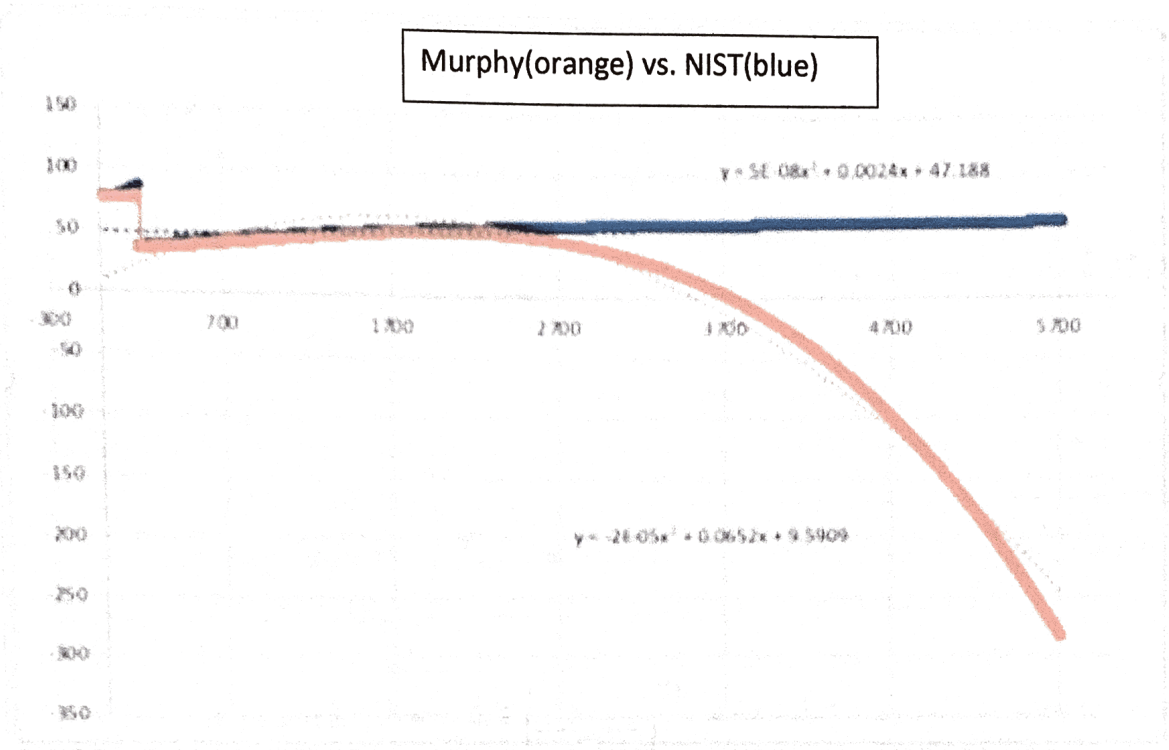
$$\Delta \hat{H} = 0 = \sum_e \dot{n}_e \Delta \hat{H}_e - \sum_i \dot{n}_i \Delta \hat{H}_i + \dot{Q} \Delta \hat{H}_r$$

$$\begin{aligned} 0 = & \left[(1 \text{ mol/s}) \left(31.32234T + \frac{(-20.23531)T^2}{2} + \frac{(57.86644)T^3}{3} + \frac{(-36.50624)T^4}{4} - \frac{(-0.007374)}{T} + (-8.9034) \right) \right. \\ & + \\ & \left[(4 \text{ mol/s}) \left(23.83491T + \frac{(12.58878)T^2}{2} + \frac{(-1.139011)T^3}{3} + \frac{(-1.497759)T^4}{4} - \frac{(0.214194)}{T} + (83.35783) \right) \right] \\ & + \\ & \left[(6 \text{ mol/s}) \left(30.09200T + \frac{(6.832514)T^2}{2} + \frac{(6.793435)T^3}{3} + \frac{(-2.534480)T^4}{4} - \frac{(0.082139)}{T} + (-250.8810) \right) \right] \\ & - \\ & \left[(4 \text{ mol/s}) (6.69866 \text{ kJ/mol}) + (6 \text{ mol/s}) (5.24836 \text{ kJ/mol}) \right] \\ & + \\ & \left[(1 \text{ mol/s}) (-905.9 \text{ kJ/mol}) \right] \end{aligned}$$

$$1055.3326 = 307.21398T + 35.537387T^2 + 31.357062T^3 - 14.42537394T^4 - \frac{1.27587}{T}$$

↳ Using Solver, we find $T_e = 2770.12 \text{ K}$

* Refer to Graph comparing Murphy vs. NIST Shomate Equations

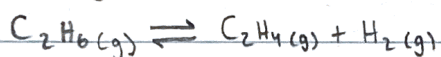


Murphy (orange) Equation: $y = 2 \times 10^{-5}x^2 + 0.0652x + 9.5909$

NIST (blue) Equation: $y = 5 \times 10^{-8}x^2 + 0.0024x + 47.188$

We only find one solution when using the Shomate equations from NIST because of the differences in the two graphs above. We see that NIST is fairly linear, giving individual, specific values for each C_p . On the other hand, Murphy begins to decrease exponentially around 2700 K, which explains why it is possible for two solutions to exist for the Shomate equation, $C_p(T)$.

2.) Dehydrogenation Reaction



$$K_p (\text{atm}) = \frac{X_{\text{C}_2\text{H}_4} X_{\text{H}_2}}{X_{\text{C}_2\text{H}_6}} P = K_0 e^{-E/T} \text{ where } K_0 = 7.28 \times 10^6 \text{ atm and } E = 17,000 \text{ K}$$

$$\Delta \hat{H}_r (@ 1273 \text{ K}) = 145.1 \text{ kJ/mol}$$

$$\left. \begin{aligned} (C_p)_{\text{C}_2\text{H}_4} &= 9.419 + 0.1147T \\ (C_p)_{\text{H}_2} &= 26.90 + 0.004167T \\ (C_p)_{\text{C}_2\text{H}_6} &= 11.35 + 0.1392T \end{aligned} \right\} T \text{ is in Kelvin}$$

$$\text{In class we proved that } f = \left(\frac{K_p}{P + K_p} \right)^{1/2} \text{ and } f = \frac{1}{1 + \phi(T)} \text{ where}$$

$$\phi(T) = \frac{\Delta \hat{H}_r (1273 \text{ K}) - \int_T^{1273 \text{ K}} [(C_p)_{\text{C}_2\text{H}_4} + (C_p)_{\text{H}_2}] dT}{\int_T^{1273 \text{ K}} [(C_p)_{\text{C}_2\text{H}_6}] dT}$$

* Combine the two results above to give an expression in the form $\phi(T) = 0$.
Then, solve the equation for different values of $P(\text{atm}) = 0.01, 0.05, 0.1, 0.5, 1.0, 5.0$
and determine the fractional conversion f . Plot T vs. P and f vs. P in 2 separate plots.

$$\phi(T) = \frac{\Delta \hat{H}_r (1273 \text{ K}) - \int_T^{1273 \text{ K}} [(C_p)_{\text{C}_2\text{H}_4} + (C_p)_{\text{H}_2}] dT}{\int_T^{1273 \text{ K}} [(C_p)_{\text{C}_2\text{H}_6}] dT}$$

$$\phi(T) = 145.1 \text{ kJ/mol} - \frac{\left[\left[9.419T + \frac{0.1147T^2}{2} \right]_T^{1273} + \left[26.90T + \frac{0.004167T^2}{2} \right]_T^{1273} \right]}{\left[11.35T + \frac{0.1392T^2}{2} \right]_T^{1273}}$$

$$\phi(T) = 145.1 - \left[(11990.387 + 92937.33815 - 9.419T - 0.05735T^2) + (34243.7 + 3376.372 - 26.90T - 0.0020835T^2) \right]$$

$$[(14448.55 + 112788.81 - 11.35T - 0.0696T^2)]$$

results from setting $f = \sqrt{\frac{K_p}{P+K_p}}$ and $f = \frac{1}{1+\phi(T)}$ equal to each other

$$f = \frac{\phi(T) = 0 = \frac{145.1 + 142547.7972 - 36.319T - 0.0594335T^2}{127237.36 - 11.35T - 0.0696T^2}}{\sqrt{\frac{K_p}{P+K_p}}}$$

(P = 0.01 atm) f =



(P = 0.05 atm) f =

(P = 0.1 atm) f =

(P = 0.5 atm) f =

(P = 1.0 atm) f =

(P = 5.0 atm) f =

Comment Summary

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good flow till here. just need type them into a computer and calculate